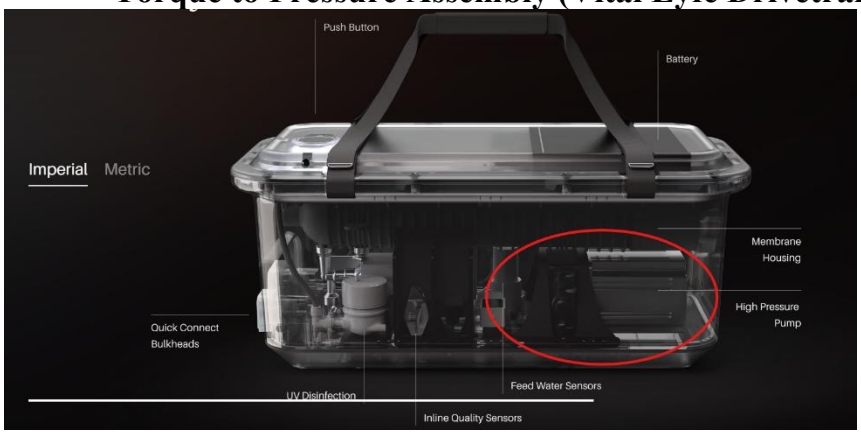
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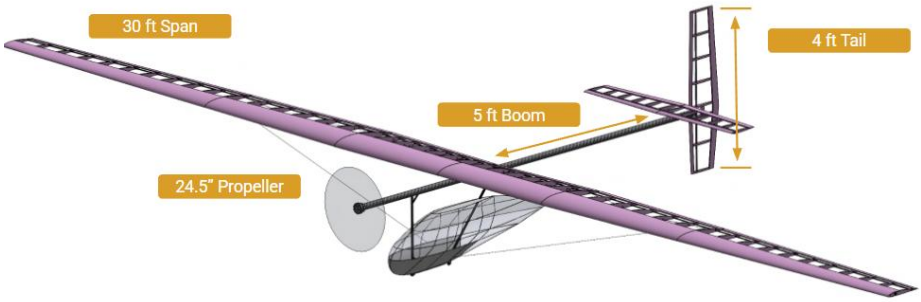
Torque to Pressure Assembly (Vital Lyfe Drivetrain)

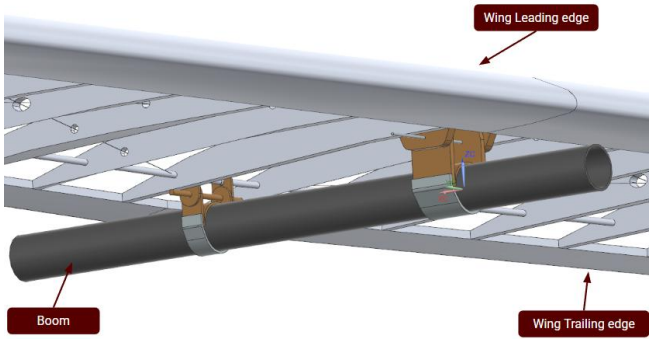




What?	How?	Results
• Take low pressure water, and push it at given flow rate and pressure requirements to send it through a membrane to clean • Size components for wire-to-water efficiency, ensure 6000-hour lifetime is maintained, design for rate manufacturing, and ensure system survives in a drop	• Size bearings, press fit, motor specs, and piston sizes via hand-calcs for first shots • Analyze cavitation and second order fluid effects by scrapping together a test rig • Define drop-load requirements to mil-spec requirements • Use COTS components, and injection mold where possible	• By far the most cost-efficient reverse osmosis system ever built • Cost and weight reduction on motor • Designed for potential upgrades to switch from metal to plastic components • Ability to achieve 120+ units per hour of manufacturability • Survivability of 40g+ drop

Hercules (HPFRT Director of Operations)





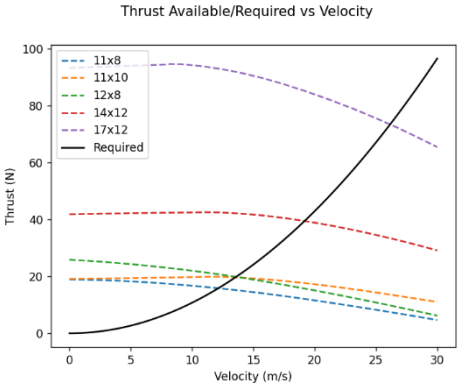
What?	How?	Results
• Design a quarter-scale plane of future human-powered aircraft • Aero design has been proven. Prove manufacturing methods including composite manufacturing and acquisition, joint design, and human-powered control systems	• Lead wing-boom integration to design a strong joint • Find a way to manufacture expensive carbon-fiber components, analyze them • Recruit new members, and assign them open projects	• Presented joint designs at a design review, hand-calced and used FEA to determine max stress • Called companies asking for sponsorships and carbon fiber, $7.5k in money raised • Increased CL max by 10%

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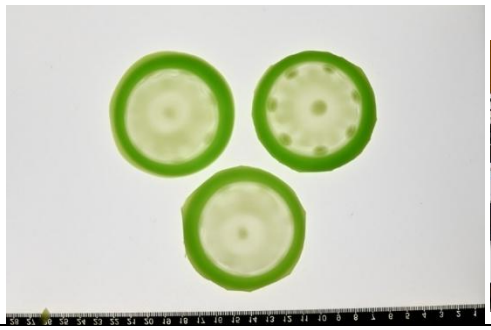
Human-powered Flight Research Team (HPFRT) Avionics Testbed



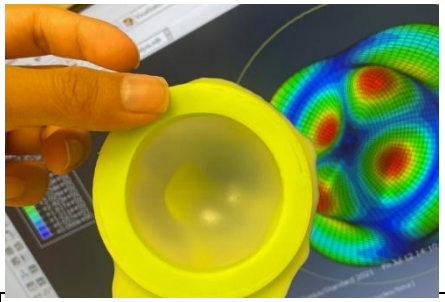


What?	How?	Results
• Design a high aspect ratio plane to serve as a technology demonstrator for avionics, flight controls, and telemetry • Emphasize stability and controllability at low Reynolds numbers, prove build methods for extreme lightweight structural members	• Perform weight build-up and estimations in Excel • Use MATLAB to conduct trade studies to do initial sizing of wing and to select airfoils • Conduct low-fidelity bending moments and stress analysis to design joints spar size	• Characterized drag via sink tests using COTS, data was published and talked about in AIAA regional conference • Achieved flight with the largest span airplane built at USC, giving go-ahead for ~30 foot airplane design

Architected Intelligent Matter Lab (AIM)







What?	How?	Results
• Design a system to accurately input small thickness imperfections on a silicon mold • Model these imperfections • Determine the corresponding structural changes	• Use sound frequencies to induce predictable thickness differences • Use Abaqus to model vibrational peaks and lows to predict imperfections on silicon • Make a syringe pressure testing system to induce buckling	• Produced over 500 silicon molds, matching frequencies predictions from Abaqus model • Determined “knock-down” factor of buckling pressure to be proportional to volume • Wrote and submitted a paper to Nature, currently under review